

Silicon

Si (Si I). Strongest classified transitions below 2 μm (NIST): 44 levels, 79 lines.

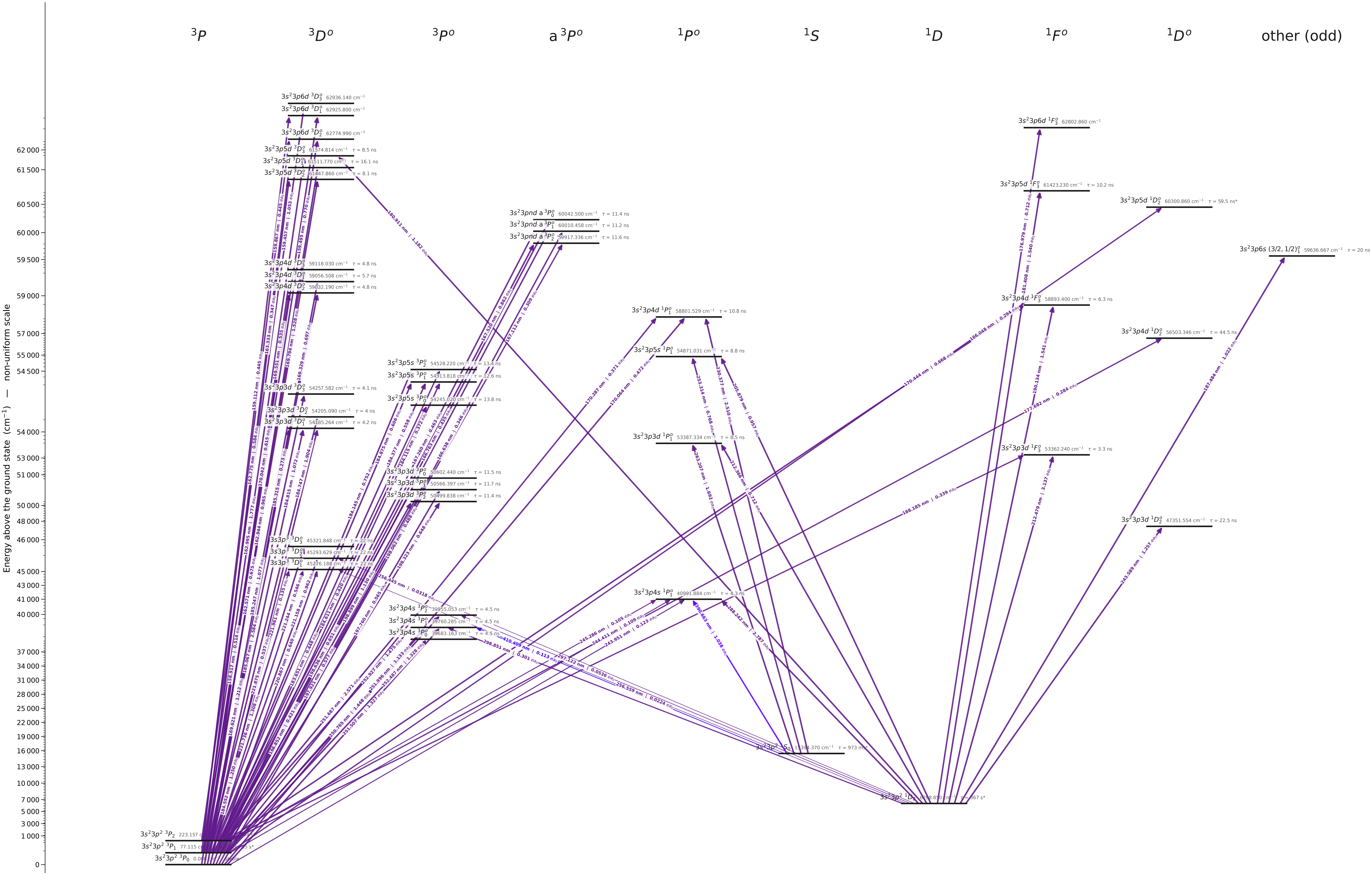
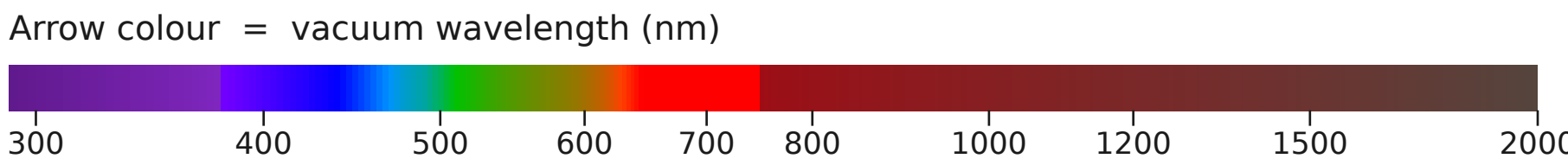
Si⁺ ionisation limit 65747.76 cm⁻¹ = 8.15168 eV (152.096 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J\rangle $ (e.a.u.)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3s^2 3p^2 \ ^3P_0 - 3s3p^3 \ ^3D_1^o$	220.8667	220.7978	1357.3460	0.640	2.570e+07	7.234	56.6	line of the 220.8-221.8 nm
$3s^2 3p^2 \ ^3P_1 - 3s3p^3 \ ^3D_2^o$	221.1581	221.0892	1355.5570	0.962	3.470e+07	7.234	76.3	line of the 220.8-221.8 nm
$3s^2 3p^2 \ ^3P_1 - 3s3p^3 \ ^3D_1^o$	221.2435	221.1745	1355.0341	0.546	1.860e+07	7.234	40.9	line of the 220.8-221.8 nm
$3s^2 3p^2 \ ^3P_2 - 3s3p^3 \ ^3D_3^o$	221.7359	221.6669	1352.0247	1.308	4.540e+07	7.234	99.9	221.74 nm
$3s^2 3p^2 \ ^3P_2 - 3s3p^3 \ ^3D_2^o$	221.8748	221.8057	1351.1788	0.537	1.070e+07	7.234	23.6	line of the 220.8-221.8 nm
$3s^2 3p^2 \ ^3P_1 - 3s^2 3p4s \ ^3P_2^o$	250.7652	250.6897	1195.5105	1.448	5.390e+07	35.37	24.3	one of the six lines of the 250.7-252.9 nm
$3s^2 3p^2 \ ^3P_0 - 3s^2 3p4s \ ^3P_1^o$	251.5073	251.4316	1191.9834	1.327	7.480e+07	35.37	33.7	one of the six lines of the 250.7-252.9 nm
$3s^2 3p^2 \ ^3P_2 - 3s^2 3p4s \ ^3P_2^o$	251.6870	251.6112	1191.1323	2.571	1.680e+08	35.37	75.7	one of the six lines of the 250.7-252.9 nm
$3s^2 3p^2 \ ^3P_1 - 3s^2 3p4s \ ^3P_1^o$	251.9960	251.9202	1189.6715	1.133	5.420e+07	35.37	24.3	one of the six lines of the 250.7-252.9 nm
$3s^2 3p^2 \ ^3P_2 - 3s^2 3p4s \ ^3P_2^o$	252.9268	252.8508	1185.2933	1.475	9.080e+07	35.37	40.9	one of the six lines of the 250.7-252.9 nm
$3s^2 3p^2 \ ^1D_2 - 3s^2 3p4s \ ^1P_1^o$	288.2423	288.1578	1040.0710	2.787	2.190e+08	37.01	94	288.158 nm
$3s^2 3p^2 \ ^1S_0 - 3s^2 3p4s \ ^1P_1^o$	390.6629	390.5523	767.3942	1.038	1.220e+07	37.01	5.2	390.552 nm
$3s^2 3p^2 \ ^1S_0 - 3s^2 3p4s \ ^3P_1^o$	410.4094	410.2936	730.4718	0.113	1.240e+05	35.37	0.056	410.294 nm

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²⁹Si hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3s^2 3p^2 \ ^3P_1$	1.840	-	F=1/2: -1.84 F=3/2: +0.92
$3s^2 3p^2 \ ^3P_2$	-160.1(13)	-	F=3/2: +240.15 F=5/2: -160.10
$3s3p \ ^3D^o_3$	-532.9(6)	-	F=5/2: +1065.80 F=7/2: -799.35

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3s^2 3p^2 \ ^3P_2 - 3s 3p^3 \ ^3D_3^o$	221.736	29-28	1753.3(11)
$3s^2 3p^2 \ ^3P_2 - 3s 3p^3 \ ^3D_3^o$	221.736	30-28	3359.9(6)
$3s^2 3p^2 \ ^3P_2 - 3s^2 3p 4s \ ^3P_2^o$	251.687	30-28	330(30)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$